



Goutham Vignesh

Applied AI Engineer | LLM Systems & Multi-Agent AI

Applied AI Engineer with 6+ years of experience building production-scale LLM, RAG, and multi-agent AI systems. Specialized in AI infrastructure, inference optimization, multimodal AI, and cloud-native ML deployment. Proven track record of delivering scalable enterprise AI solutions with measurable business impact.

 gouthamvigneshs@gmail.com  +91 9952689236  Bangalore, India

WORK EXPERIENCE

01/2024 - Present	AI RESEARCH ENGINEER - APPLIED R&D UNIT - Developed a multi-agent AI framework using Google ADK, MCP, and LangGraph for scalable tool orchestration, reusable skills, and enterprise workflow automation. - Led ML engineering for the IQVIA engagement by designing LangGraph-based workflows to automate clinical trial data review and accelerate delivery timelines. - Built AI coding assistants for Quantiphi's Codeaira platform with agent orchestration and skill-planning capabilities, reducing manual effort by 60%. - Developed IDPFlow, a no-code multi-agent framework for reusable document AI workflows, improving processing efficiency by 40%. - Deployed IDPFlow on Gemini Enterprise as an Agent-as-a-Service platform for scalable document intelligence and workflow automation.
04/2023 - 01/2024	RESEARCH ENGINEER (GEN AI) - Built cost-efficient LLM fine-tuning pipelines using PEFT and LoRA, reducing training costs by 75%. - Optimized open-source LLM deployment using vLLM, NVIDIA Triton, and quantization techniques to improve inference scalability and latency. - Applied model compression and quantization strategies for efficient LLM deployment on resource-constrained infrastructure.
12/2019 - 03/2023	MACHINE LEARNING ENGINEER - Built RAG-based document retrieval and semantic search systems using Pinecone and Weaviate for legal document intelligence. - Developed intelligent document processing pipelines combining OCR, OpenCV, and YOLO-based object detection for document classification and information extraction, achieving 85% accuracy in address-proof verification workflows. - Built scalable ML deployment pipelines and APIs for production model serving, improving deployment efficiency and system reliability by 30%.

PUBLICATIONS

- **IDPFlow:** *A No-Code Framework for Multimodal Intelligent Document Processing* — ACM Multimedia 2025
- **DocAnnot:** *Accelerating Key Information Extraction Dataset Creation* — Book Chapter, 2026
- **DocGenie:** *A Framework for High-Fidelity Synthetic Document Generation* — CVPR Workshop 2025
- **SciSpace Copilot:** *Intelligent Reading Assistance for Researchers* — AAAI 2024
- **SciSpace Literature Review:** *AI for Scientific Discovery* — ECIR 2024

SKILLS

Programming & Frameworks

- Python, FastAPI, REST APIs, Git, Linux

AI & Machine Learning

- PyTorch, TensorFlow, Transformers, Reinforcement Learning, Ensemble Learning, Computer Vision

LLM & Agentic Systems

- LLMs, RAG, LangGraph, MCP, Google ADK, Multi-Agent Systems, PEFT, LoRA

Multimodal & Document AI

- OCR, Document Intelligence, OpenCV, YOLO, Semantic Search, Knowledge Graphs

Deployment & Infrastructure

- Docker, Kubernetes, NVIDIA Triton, vLLM, AWS, GCP, Azure

Databases & Vector Stores

- MongoDB, Pinecone, Weaviate

EDUCATION

2020	Bachelor of Engineering
Sri Krishna College of Technology, affiliated to Anna University Coimbatore, India	Computer Science Engineering

LANGUAGES

- **English - Professional Working Proficiency**

WEBSITES

- [LinkedIn](#) | [GitHub](#) | [Google Scholar](#) | [Portfolio](#) | [ORCID](#)